

Theorem and Corollary Summary of the Lean Proof for RandNLA Algorithm 1 Stability

LeanFpAnalysis repository

Overview

This note summarizes the Lean formalization of the entrywise floating-point stability analysis for Algorithm 1, the element-wise sampling meta-algorithm from the RandNLA paper. The formalized sampling probabilities are the squared-magnitude probabilities

$$p_{ij} = \frac{A_{ij}^2}{\sum_{k,l} A_{kl}^2} = \frac{A_{ij}^2}{\|A\|_F^2}.$$

The document is written as mathematical theorem and corollary statements. Each statement also records the corresponding Lean theorem or definition name.

Notation and assumptions

Let $A \in \mathbb{R}^{m \times n}$, let s be the sampling parameter from Algorithm 1, and fix an entry (i, j) . Let $\tilde{A}^{(0)}$ be the initial sketch and $\hat{A}$ be the floating-point sketch produced by the algorithm.

The floating-point model is the repository's standard relative-error model: for each elementary operation,

$$\text{fl}(x \circ y) = (x \circ y)(1 + \delta), \quad |\delta| \leq u,$$

where u is the unit roundoff. The accumulated-error notation is Higham's

$$\gamma_q = \frac{qu}{1 - qu}.$$

Lean represents the required side condition $qu < 1$ as `gammaValid fp q`.

Throughout the main entrywise theorem, the relevant assumptions are:

$$s \neq 0, \quad A_{ij} \neq 0, \quad \text{gammaValid fp } q_{\{ij\}}, \quad \text{gammaValid fp } (q_{\{ij\}} + 1).$$

Deterministic arithmetic

Definition 1 (Squared-magnitude increment). For a sampled nonzero entry (i, j) , Algorithm 1 adds

$$\frac{A_{ij}}{sp_{ij}}$$

to the sketch entry. Under squared-magnitude probabilities this increment is

$$\frac{A_{ij}}{sp_{ij}} = \frac{\|A\|_F^2}{sA_{ij}}.$$

Lean reference. Formalized as `elementwiseIncrement_sqMag_eq`.

Theorem 2 (One-sample floating-point stability). *Fix a scalar sketch value x . Suppose $s \neq 0$ and $A_{ij} \neq 0$. The floating-point update*

$$\text{fl} \left(x + \text{fl} \left(\frac{A_{ij}}{s p_{ij}} \right) \right)$$

satisfies

$$\begin{aligned} & \left| \hat{x} - \left(x + \frac{\|A\|_F^2}{s A_{ij}} \right) \right| \\ & \leq \left| x + \frac{\|A\|_F^2}{s A_{ij}} \right| u + \left| \frac{\|A\|_F^2}{s A_{ij}} \right| u(1 + u). \end{aligned}$$

Lean reference. Formalized as `fl_elementwiseUpdateEntry_sqMag_error_bound`.

Trace reduction

Definition 3 (Hit counter). For a deterministic trace

`samples : Fin steps -> Fin m x Fin n,`

the hit counter q_{ij} is the number of trace steps that sample entry (i, j) :

$$q_{ij} = \#\{t : \text{Fin steps} \mid \text{samples}(t) = (i, j)\}.$$

Lean reference. Formalized as `hitCount samples i j`.

Lemma 4 (Trace equals repeated scalar updates). *For a fixed output entry (i, j) , the exact trace and the floating-point trace depend on the full sample trace only through q_{ij} . In other words, the entrywise trace update is equivalent to repeating the corresponding scalar entry update exactly q_{ij} times.*

Lean reference. Formalized as

`elementwiseTraceSketch_eq_repeat_of_hitCount,`
`fl_elementwiseTraceSketch_eq_repeat_of_hitCount.`

Main deterministic result

Theorem 5 (Deterministic entrywise stability for Algorithm 1). *Fix an entry (i, j) , and let q_{ij} be its hit count in the trace. Under the assumptions above, Algorithm 1 with squared-magnitude sampling satisfies*

$$\begin{aligned} & \left| \hat{\tilde{A}}_{ij} - \left(\tilde{A}_{ij}^{(0)} + q_{ij} \frac{\|A\|_F^2}{s A_{ij}} \right) \right| \\ & \leq \left| \tilde{A}_{ij}^{(0)} \right| \gamma_{q_{ij}} + q_{ij} \left| \frac{\|A\|_F^2}{s A_{ij}} \right| \gamma_{q_{ij}+1}. \end{aligned}$$

Lean reference. Formalized as `fl_elementwiseTraceSketch_sqMag_error_bound`.

Corollary 6 (Comparison with the exact trace). *The same deterministic bound can be written as a direct comparison between the floating-point trace and the exact arithmetic trace produced by the same sample sequence.*

Lean reference. Formalized as `fl_elementwiseTraceSketch_sqMag_error_bound_exact`.

Corollary 7 (Zero-initial-sketch Algorithm 1). *If Algorithm 1 starts from the zero sketch, so that $\tilde{A}_{ij}^{(0)} = 0$, then*

$$\left| \hat{A}_{ij} - q_{ij} \frac{\|A\|_F^2}{sA_{ij}} \right| \leq q_{ij} \left| \frac{\|A\|_F^2}{sA_{ij}} \right| \gamma_{q_{ij}+1}.$$

Lean reference. Formalized as `fl_elementwiseSketch_zero_init_sqMag_error_bound`.

Deterministic budget from a count bound

Definition 8 (Stability budget). For a deterministic upper bound Q on the hit counter, define

$$B(Q) = \left| \tilde{A}_{ij}^{(0)} \right| \gamma_Q + Q \left| \frac{\|A\|_F^2}{sA_{ij}} \right| \gamma_{Q+1}.$$

Lean reference. Formalized as `sqMagTraceErrorBudget`.

Lemma 9 (Budget monotonicity). *If $q_{ij} \leq Q$, then the deterministic stability budget at q_{ij} is bounded by the deterministic stability budget at Q :*

$$B(q_{ij}) \leq B(Q).$$

Lean reference. Formalized as `sqMagTraceErrorBudget_mono`.

Corollary 10 (Stability from a deterministic count bound). *If $q_{ij} \leq Q$, then Algorithm 1 satisfies the deterministic entrywise stability bound with $B(Q)$ in place of the trace-specific budget $B(q_{ij})$.*

Lean reference. Formalized as

`fl_elementwiseTraceSketch_sqMag_error_bound_of_hitCount_le`.

Random trace model

Definition 11 (Random trace). Let X be a random trace in a finite probability space. For fixed (i, j) , the random variable of interest is

$$q_{ij}(X) = \text{hitCount } (X \text{ omega}) \text{ i j}.$$

For fixed A , s , floating-point model, and initial sketch, this hit counter is the only random quantity in the deterministic stability budget.

Lean reference. The finite probability interface is `FiniteProbability`, and the identity writing the hit count as a sum of indicators is `hitCount_eq_sum_indicator`.

Theorem 12 (Mean hit count under the marginal sampling law). *Assume that each sample step hits entry (i, j) with marginal probability p_{ij} . Then*

$$\mathbb{E}[q_{ij}] = \text{steps} \cdot p_{ij}.$$

For squared-magnitude sampling,

$$p_{ij} = \frac{A_{ij}^2}{\|A\|_F^2}.$$

Lean reference. Formalized as

`expectationNat_hitCount_eq_steps_mul_hitProb` and its squared-magnitude specialization.

Markov high-probability result

Theorem 13 (Markov concentration for the hit counter). *Assume only the marginal law*

$$\mathbb{P}(\text{step } t \text{ hits } (i, j)) = p_{ij}$$

for every step t . Then, for any natural number Q ,

$$\mathbb{P}(q_{ij} \leq Q) \geq 1 - \frac{\text{steps} \cdot p_{ij}}{Q + 1}.$$

Equivalently, if

$$\frac{\text{steps} \cdot p_{ij}}{Q + 1} \leq \delta,$$

then

$$\mathbb{P}(q_{ij} \leq Q) \geq 1 - \delta.$$

Lean reference. Formalized as

`hitCount_concentration_sqMag_markov` and `hitCount_concentrates_sqMag`.

Corollary 14 (Chosen Markov budget). *Let $\delta > 0$, and choose*

$$Q_M = \left\lceil \frac{\text{steps} \cdot p_{ij}}{\delta} \right\rceil.$$

Then

$$\mathbb{P}(q_{ij} \leq Q_M) \geq 1 - \delta.$$

Lean reference. The chosen count budget is formalized as `sqMagMarkovHitCountBudget`. The tail inequality is formalized as `sqMagMarkovHitCountBudget_tail`.

Corollary 15 (High-probability stability with the Markov budget). *Under the squared-magnitude marginal sampling law, with Q_M as above, Algorithm 1 satisfies the deterministic stability bound with budget $B(Q_M)$ with probability at least $1 - \delta$:*

$$\mathbb{P} \left(\left| \widehat{A}_{ij} - \left(\widetilde{A}_{ij}^{(0)} + q_{ij} \frac{\|A\|_F^2}{sA_{ij}} \right) \right| \leq B(Q_M) \right) \geq 1 - \delta.$$

Lean reference. Formalized as `highProbability_sqMagTraceStability_of_markov_budget`.

Pairwise-Chebyshev high-probability result

Theorem 16 (Centered second moment under pairwise independence). *Assume the marginal law above and assume that distinct step-hit indicators for entry (i, j) are pairwise independent. Then Lean proves an explicit formula for*

$$M = \mathbb{E} \left[(q_{ij} - \text{steps} \cdot p_{ij})^2 \right].$$

This is the variance-scale quantity used by Chebyshev's inequality.

Lean reference. The centered moment is formalized as `hitCountPairwiseCenteredMoment`. The theorem identifying it with the expectation is

$$\text{expectationReal_hitCount_centered_sq_eq_pairwise.}$$

Theorem 17 (Chebyshev concentration around the mean). *Under the same marginal and pairwise-independence assumptions, if*

$$\frac{M}{\varepsilon^2} \leq \delta,$$

then

$$\mathbb{P}(|q_{ij} - \text{steps} \cdot p_{ij}| \leq \varepsilon) \geq 1 - \delta.$$

For squared-magnitude sampling, $p_{ij} = A_{ij}^2 / \|A\|_F^2$.

Lean reference. Formalized as `hitCount_concentrates_sqMag_around_mean_pairwise`.

Corollary 18 (Chosen Chebyshev budget). *Let $\delta > 0$, let*

$$M = \mathbb{E} \left[(q_{ij} - \text{steps} \cdot p_{ij})^2 \right],$$

and choose the conservative radius

$$\varepsilon_C = \frac{M}{\delta} + 1.$$

Then Lean proves

$$\frac{M}{\varepsilon_C^2} \leq \delta.$$

Consequently, if

$$Q_C = \lceil \text{steps} \cdot p_{ij} + \varepsilon_C \rceil,$$

then

$$\mathbb{P}(q_{ij} \leq Q_C) \geq 1 - \delta.$$

Lean reference. The chosen radius and budget are formalized as

$$\text{chebyshevHitCountRadius, sqMagChebyshevHitCountBudget.}$$

The key arithmetic lemmas are

$$\text{hitCountPairwiseCenteredMoment_div_chebyshevRadius_sq_le, sqMagChebyshevHitCountBudget_mean_add_radius_le.}$$

Corollary 19 (High-probability stability with the Chebyshev budget). *Under the squared-magnitude marginal law and pairwise independence of distinct step-hit indicators, with Q_C as above, Algorithm 1 satisfies the deterministic stability bound with budget $B(Q_C)$ with probability at least $1 - \delta$:*

$$\mathbb{P} \left(\left| \widehat{A}_{ij} - \left(\widetilde{A}_{ij}^{(0)} + q_{ij} \frac{\|A\|_F^2}{sA_{ij}} \right) \right| \leq B(Q_C) \right) \geq 1 - \delta.$$

Lean reference. Formalized as

$$\text{highProbability_sqMagTraceStability_of_pairwise_chebyshev_budget.}$$

Chernoff high-probability result

Theorem 20 (Canonical product-sampler MGF). *Assume $\|A\|_F^2 > 0$, and let P_A be the finite product probability law that samples every trace step independently with probability*

$$p_{kl} = \frac{A_{kl}^2}{\|A\|_F^2}.$$

Then, for every λ ,

$$\mathbb{E}_{P_A}[\exp(\lambda q_{ij})] = (1 + p_{ij}(\exp(\lambda) - 1))^{\text{steps}}.$$

In particular, for $\lambda > 0$,

$$\mathbb{E}_{P_A}[\exp(\lambda q_{ij})] \leq \exp(\text{steps} \cdot p_{ij} \cdot (\exp(\lambda) - 1)).$$

Lean reference. The product law is `sqMagTraceProbability`. The exact MGF identity is

$$\text{sqMagTraceProbability_expectationReal_exp_hitCount_eq},$$

and the derived Chernoff MGF bound is

$$\text{sqMagTraceProbability_chernoff_mgf_bound}.$$

Corollary 21 (Chernoff concentration for the canonical sampler). *Under the same canonical independent sampling law, for every natural number Q and every $\lambda > 0$,*

$$\mathbb{P}_{P_A}(q_{ij} \leq Q) \geq 1 - \exp(\text{steps} \cdot p_{ij} \cdot (\exp(\lambda) - 1) - \lambda(Q + 1)).$$

Lean reference. The generic MGF condition is still named `sqMagHitCountChernoffMGFBound`, but the canonical-sampler theorem uses the proved product-law MGF internally:

$$\text{hitCount_concentrates_sqMag_chernoff_independent}.$$

Corollary 22 (Chosen fixed-parameter Chernoff budget). *Let $\delta > 0$ and $\lambda > 0$. For the canonical independent sampler, choose*

$$Q_{\text{Ch}}(\lambda) = \left\lceil \frac{\text{steps} \cdot p_{ij} \cdot (\exp(\lambda) - 1) - \log(\delta)}{\lambda} \right\rceil.$$

Then the Chernoff tail is at most δ , and hence

$$\mathbb{P}_{P_A}(q_{ij} \leq Q_{\text{Ch}}(\lambda)) \geq 1 - \delta.$$

Lean reference. The chosen budget is `sqMagChernoffHitCountBudget`, and the tail inequality is `sqMagChernoffHitCountBudget_tail`. The fully derived canonical-sampler concentration statement is

$$\text{hitCount_concentrates_sqMag_chernoff_independent_budget}.$$

Corollary 23 (Canonical high-probability stability with the Chernoff budget). *For the canonical independent squared-magnitude sampler, Algorithm 1 satisfies the deterministic stability bound with budget $B(Q_{\text{Ch}}(\lambda))$ with probability at least $1 - \delta$. That is,*

$$\mathbb{P}_{P_A} \left(\left| \tilde{A}_{ij} - \left(\tilde{A}_{ij}^{(0)} + q_{ij} \frac{\|A\|_F^2}{sA_{ij}} \right) \right| \leq B(Q_{\text{Ch}}(\lambda)) \right) \geq 1 - \delta.$$

Lean reference. Formalized as

`highProbability_sqMagTraceStability_of_independent_bernoulli_budget.`

Theorem 24 (Optimized Chernoff exponent for the canonical sampler). *For a supplied natural budget Q , assume*

$$0 < \mu = \text{steps} \cdot p_{ij} \quad \text{and} \quad \mu < Q + 1.$$

Lean chooses the optimizing parameter

$$\lambda_* = \log \left(\frac{Q + 1}{\mu} \right)$$

and proves the corresponding optimized Chernoff tail. Thus, if

$$\text{sqMagChernoffOptimizedHitCountTail}(Q) \leq \delta,$$

then

$$\mathbb{P}_{P_A}(q_{ij} \leq Q) \geq 1 - \delta.$$

Lean reference. Formalized as

`hitCount_concentrates_sqMag_bernoulli_optimized_independent,`
`highProbability_sqMagTraceStability_of_independent_bernoulli_optimized_tail_budget.`

Generic probability-to-stability transfer

Theorem 25 (Transferring hit-count concentration to stability). *Let Q be any deterministic natural number. If*

$$\mathbb{P}(q_{ij} \leq Q) \geq 1 - \delta,$$

then the Algorithm 1 stability event with deterministic budget $B(Q)$ also has probability at least $1 - \delta$:

$$\mathbb{P}(\text{stability bound with budget } B(Q)) \geq 1 - \delta.$$

Lean reference. Formalized as

`highProbability_sqMagTraceStability_of_hitCount_concentration.`

Corollary 26 (Tail-budget version). *If the Markov tail condition*

$$\frac{\text{steps} \cdot p_{ij}}{Q + 1} \leq \delta$$

holds, then the stability event with budget $B(Q)$ holds with probability at least $1 - \delta$.

Lean reference. Formalized as

`highProbability_sqMagTraceStability_of_marginal_hitProb_of_tail_budget.`

Corollary 27 (Deviation-budget version). *If Chebyshev gives*

$$\mathbb{P}(|q_{ij} - \text{steps} \cdot p_{ij}| \leq \varepsilon) \geq 1 - \delta$$

and the chosen natural budget satisfies

$$\text{steps} \cdot p_{ij} + \varepsilon \leq Q,$$

then the stability event with budget $B(Q)$ holds with probability at least $1 - \delta$.

Lean reference. Formalized as

`highProbability_sqMagTraceStability_of_pairwise_hitCount_deviation.`

What is, and is not, proved

The formalization proves entrywise floating-point stability for Algorithm 1, first deterministically and then with high probability by proving explicit concentration bounds for the random hit counter q_{ij} .

The Markov corollary needs only marginal hit probabilities, but it is crude. The Chebyshev corollary uses pairwise independence and proves concentration around the mean before transferring that result into the floating-point stability bound. The Chernoff corollary constructs the canonical independent squared-magnitude product law for Algorithm 1 traces, proves the hit-count MGF bound from that law, and gives the exponentially sharper tail used in the tightest counter-level analysis.

The current formalization does not yet prove matrix-level RandNLA accuracy results such as spectral-norm concentration, unbiasedness of the full sketch, or sample-complexity theorems.

Lean artifact table

Mathematical object or result	Lean artifact
Rectangular Frobenius norm	frobNormSqRect
Squared-magnitude probabilities	sqMagProb
Squared-magnitude increment identity	elementwiseIncrement_sqMag_eq
One-sample stability	fl_elementwiseUpdateEntry _sqMag_error_bound
Trace hit count	hitCount
Deterministic budget	sqMagTraceErrorBudget
Canonical product trace law	sqMagTraceProbability
Canonical Chernoff MGF theorem	sqMagTraceProbability _chernoff_mgf_bound
Exact trace reduction	elementwiseTraceSketch _eq_repeat_of_hitCount
Floating-point trace reduction	fl_elementwiseTraceSketch _eq_repeat_of_hitCount
Main deterministic theorem	fl_elementwiseTraceSketch _sqMag_error_bound
Zero-initial corollary	fl_elementwiseSketch _zero_init_sqMag_error_bound
Markov chosen budget	sqMagMarkovHitCountBudget
Markov stability corollary	highProbability_sqMagTraceStability _of_markov_budget
Chebyshev chosen budget	sqMagChebyshevHitCountBudget
Chebyshev stability corollary	highProbability_sqMagTraceStability _of_pairwise_chebyshev_budget
Chernoff chosen budget	sqMagChernoffHitCountBudget
Optimized Chernoff tail	sqMagChernoffOptimizedHitCountTail
Chernoff stability corollary	highProbability_sqMagTraceStability _of_independent_chernoff_budget
Optimized Chernoff stability	highProbability_sqMagTraceStability _of_independent_chernoff _optimized_tail_budget

The main source files are:

LeanFpAnalysis/FP/Algorithms/RandNLA/ElementwiseSampling.lean

and

LeanFpAnalysis/FP/Algorithms/RandNLA/HitCountConcentration.lean.